## PHASE 1: OFFLINE TRAINING

**Training Partition $\mathcal{D}_{\text{train}}$ (80%)**

Stratified 80/10/10 Splits
Benchmark Cross-Evaluation

**Stage 1: Primary GBDT Core**

Cost-Weighted Ensemble $f_{\text{core}}(\mathbf{x})$
Balanced Weighting ($w_{\text{pos}}$)

Validation Score

## PHASE 2: VALIDATION & CALIBRATION

**Validation Partition $\mathcal{D}_{\text{val}}$ (10%)**

Validation Inference: $\hat{p}_{\text{val}} = f_{\text{core}}(\mathbf{x})$
Held-Out Tuning Partition

**Stage 2: Optimal Threshold $\tau^*$ Search**

$\tau^* = \arg\min_{\tau} \text{Cost}(\tau)$
Uniform Grid Search $\tau \in [0.01, 0.99]$

$\hat{p} \geq \tau^*$

**5-Fold OOF Cross-Fitting**

Candidate Alerts $\mathcal{D}_{\text{cand}}$ ($\hat{p} \geq \tau^*$)
TreeSHAP $\phi(\mathbf{x})$ + Meta-Model $g^{(j)}$

OOF Meta-Risk $\hat{q}$

**Amount-Tiered Safety Calibration**

Quantile Stratification ($K = 3$ Tiers)
Safety Thresholds $\tau_{\text{meta},k}$ ($\Delta\text{FN}_{\text{val}} = 0$)

**PRODUCTION DEPLOYMENT PIPELINE (Algorithm 1) : Deploy $\{f_{\text{core}}, \tau^*, g^{(j)}, \{\tau_{\text{meta},k}\}\} \longrightarrow$ Streaming Runtime Engine**

Deploy $f_{\text{core}}$ & $\tau^*$

Deploy $g$ & $\{\tau_{\text{meta},k}\}$

## PHASE 3: STREAMING PRODUCTION DEPLOYMENT WITH SAFETY GUARD

**Live Transaction**

Feature Vector $\mathbf{x} \in \mathbb{R}^d$
Real-Time Ingestion
Single-Row ($B = 1$)

**Stage 1 Fast-Path**

Score $\hat{p} = f_{\text{core}}(\mathbf{x})$
Sub-millisecond Scoring
Gate: $\hat{p} \geq \tau^*$ ?

$\hat{p} < \tau^*$

**Auto-Cleared Legit**

Direct Approval ($\hat{y} = 0$)

$\hat{p} \geq \tau^*$

**Stage 3 TreeSHAP**

Local Attribution $\phi(\mathbf{x})$
Exact TreeExplainer
Selective Trigger

**Meta-Classification**

Joint Input $[\mathbf{x}, \phi(\mathbf{x})]$
Ensemble $g(\mathbf{u}) \to \bar{q}$
Secondary Filter

**Safety Guard**

Tier $k$(Amount)
$\bar{q} \geq \tau_{\text{meta},k}$ ?

$\bar{q} \geq \tau_{\text{meta},k}$

**Auto-Cleared False Alarm ($\hat{y} = 0$)**

$\bar{q} < \tau_{\text{meta},k}$

**Escalated Candidate Alert ($\hat{y} = 1$)**